

MTH203: Multivariate Calculus

Monsoon 2025

Name: _____

Roll No.: _____

Tutorial Quiz 4

Question:

1. Sketch the region of integration and evaluate the integral:

$$\int_0^{\pi} \int_0^{\sin x} y \, dy dx$$

(1)

2. Find the centroid of the region in the first quadrant that is bounded above by the line $y = x$ and below the parabola $y = x^2$.

(2)

Please note: There is no partial marking. Only integer marking.